

Optimized Solutions Limited



Certificate of Project Completion (Final)

Date : 12/09/2024
Project Title : IOT BASED MONITORING OF THE WATER FLOW METER
Project Number : 433
Name of Organization : HINDUSTAN AERONAUTICS LIMITED (HAL) - KORAPUT
Internal Work Order No : OS/IWO/0433/23-24

This is to certify that the task assigned to Optimized Solutions Ltd with the above purchase/work order has been successfully completed.

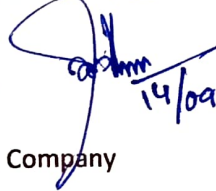
Name:

V. S. Pani

Designation:

CM (Works), HAL Sunabeda.

Signature:


14/09/2024

Seal of the Company

Date:

For Office Use Only:

Approved by: PRAJAPATI SAHIL

Signature: 

Designation: Automation Engineer.

Date: 14/09/2024.

MINUTES OF MEETING (WITH CUSTOMER)

Application Engineer's Name: Sahil Prajapati

Date: 14/09/2024

Company Name	Hindustan Aeronautics Limited(HAL)			
Company Address	PO Sunabeda, 763002, Di. Koraput, Odisha			
Ref No. If Any	PURCHASE ORDER NO: CCW/6105/24			
Customer Name	Designation	From OSL Name	Designation	
1. U.S. Pare	CN (Nargo)	1. Sahil Prajapati	Automation Engineer	
2. L N Gouda.	CN (Hcart.)	2. Devansh Nirmal	Jr. Software Engineer.	
Email ID: civil-sed.koraput@hal-india.co.in		Contact No:		
Website		Mobile No: 7656966330		
Industry & Product				

Agenda

Supply, Installation and commissioning of IoT based monitoring of the water flow meters.

Customer Requirement

Supply, Installation and commissioning of IoT based monitoring of the water flow meters.

5. Vendor to supply & fix 4G Gateway, duly configured.

Discussions & Decisions - Vendor to ensure smooth flow transition to NIC, To confirm.

Engineers from OSL started working on 5th September and completed on 12th Sep 2024 and the detailed work progress has been given below:

1. OSL installed their IOT system as per given location by HAL.
2. OSL have successfully completed all the final testing with Flow meters as per Purchase order and the requirement of HAL.
3. As per request, current SIM card is supplied by OSL on temporary basis for this month. However, New SIM card will be arranged by HAL soon for IOT data access.
4. The server runs through the command line interface on the host PC, and if the command line window is closed, the server will stop and data will be lost. Additionally, the application needs continuous internet access (24/7) to effectively capture and transmit data to the NIC server. Without constant internet connectivity, the application will not be able to perform these functions.

5.

Prepared by
Sahil Prajapati
14/09/24

14/09/24
(L N Gouda)
CN (Hcart.)

14/09/24
U.S. Pare
CN (Nargo)

Approved By



To Whomsoever It May Concern

HINDUSTAN AERONAUTICS LIMITED (HAL) – KORAPUT expressed their requirement for IoT based monitoring of the water flow meters consisting of meters to efficiently monitor water consumption on the real-time basis.

We are extremely satisfied by Digital Transformation Services of Optimized Solutions Ltd (OSL) against PO number **CCW/6105/24** dated **16/02/2024** for the development of **“IoT Based Monitoring of the Water Flow Meters”**.

Hal Koraput would like to express our deepest appreciation to Optimized Solutions for their exceptional partnership throughout the successful execution of our project. Their expertise in hardware procurement and comprehensive end-to-end development, testing, and deployment services was instrumental in delivering a seamless, high-quality solution.

From the outset, Optimized Solutions demonstrated a deep understanding of our project's functional requirements by thoroughly defining the scope, developing functional specifications, and meticulously reviewing the FRD (Functional Requirements Document) with us. Their team ensured that all of our feedback was incorporated, and they achieved full alignment with our goals before moving forward. Their proficiency in wireframing was equally impressive, producing clear and user-friendly designs that reflected the project's specifications. These wireframes underwent multiple reviews, with feedback from our team thoughtfully integrated before final approval. Through exemplary project management, Optimized Solutions maintained a well-structured project plan. They defined and executed sprints with precision, conducting regular scrums and providing detailed reports that allowed us to stay informed about every stage of development. Their attention to planning and communication was pivotal in keeping the project on track.

On the design front, Optimized Solutions successfully delivered a robust system architecture while finalizing the tools and technologies required. They designed intuitive and responsive web applications, aligning branding and layouts with our specifications, and delivered beautifully crafted UI designs using HTML and CSS that exceeded our expectations. Their thorough approach to database implementation ensured that all object properties were accurately defined, with appropriate relationships, constraints, and indexes in place. This careful planning enabled smooth application performance and data flow.

Throughout the application development phase, Optimized Solutions not only ensured our solution was built on a solid foundation, but also delivered a tailored IoT-driven water flow monitoring system to meet Hal Koraput's specific requirements. The system integrated



multiple water meters to track and monitor water consumption efficiently. This solution involved live data monitoring where meter data was stored on HAL's servers for real-time access. Optimized Solutions implemented a dashboard that provided immediate access to live data, allowing easy monitoring of water usage and cost evaluation, with seamless integration into the government portal for transparent reporting. The development team also handled key features such as user login, live data monitoring, and customizable reports, ensuring smooth operation and maximum utility.

Optimized Solutions also provided comprehensive QA testing, where their meticulously designed test plans and strategies ensured the robustness of our solution. The QA environment, combined with rigorous unit and integrated testing, guaranteed a reliable and responsive user experience. In terms of deployment and support, Optimized Solutions handled both UAT/staging and production environments with the utmost professionalism. Their on-premises deployment configurations, setup of project pre-requisites, and code submissions were executed with flawless precision.

Hal Koraput extends our heartfelt thanks to Optimized Solutions Digital Transformation, Monitoring & Automation with Advanced Data Analytics team for their commitment to excellence at every stage of this project. Their role in providing high-quality hardware, coupled with outstanding development, testing, and deployment services, has been vital to our success. We look forward to continuing this productive collaboration and achieving further milestones together.

For HINDUSTAN AERONAUTICS LIMITED (HAL) - KORAPUT


Signature

Name *L N Girda, . . .*
CM (Mast) ~